

Workshop: Statistical frequency analysis and data visualization

Exercise 1: Customer Segmentation & Qualitative Frequencies

Core Topic: Business Intelligence & MIS Visualization

Objective: To process qualitative (categorical) business data into a frequency table and visualize it for a Management Information System (MIS) dashboard using both MS Excel and R.

Part 1: The Business Scenario

Company: OmniRetail Global

Problem: The Marketing Director needs an MIS dashboard showing a breakdown of customer preferences (e.g., preferred payment methods, product categories, or shipping choices).

Your Role: Junior Data Analyst.

Generating the dataset

Run the following R script to generate your own "OmniRetail Global" data.

```
# OmniRetail Global: Data Generation Script
# Run this script to create your 2000-record dataset for the workshop
# Set seed for reproducibility
set.seed(123)

# 1. Define the sample size
n <- 2000

# 2. Generate Qualitative Data: Payment Types
# Simulating a realistic distribution of payment preferences
payment_types <- c("Credit Card", "Digital Wallet", "Debit Card", "Bank Transfer",
"Cryptocurrency")
Payment_Type <- sample(payment_types, n, replace = TRUE, prob = c(0.40, 0.30, 0.15, 0.10, 0.05))

# 3. Generate Quantitative Data: Transaction Processing Times (in milliseconds)
# Simulating a legacy system with a right-skewed distribution (mostly fast, some very slow)
Processing_Time <- round(rlnorm(n, meanlog = 4.8, sdlog = 0.5))

# 4. Combine into a dataframe
omni_data <- data.frame(Payment_Type, Processing_Time)

# 5. Export to CSV
write.csv(omni_data, "omniretail_simulated_data.csv", row.names = FALSE)
print("Dataset 'omniretail_simulated_data.csv' generated successfully!")
```

Part 2: Exercise Solutions

Exercise 1: Customer Segmentation & Qualitative Frequencies

MS Excel Solution:

1. Frequency: Open omniretail_simulated_data.csv in Excel. Create a summary table with the unique payment types. In the adjacent column, use the formula =COUNTIF(A:A, "Credit Card") (adjusting the criteria for each type) to calculate the frequency.
2. Relative Frequency: Sum the total frequencies (which should equal 2000). In a third column, divide each individual frequency by the total (n=2000) and format the cells as percentages.
3. Visualization: Highlight the summary table (Payment Type and Frequency). Go to Insert and select a standard Pie Chart and a 2D Bar Chart to build the MIS dashboard components.

R Programming Solution:

```
# 1. Read the simulated CSV file
data = read.table('omniretail_simulated_data.csv', header = TRUE, sep = ",")
# 2. Calculate the frequency table for Payment_Type
freq <- table(data$Payment_Type)
print(freq)
# 3. Calculate and print relative percentage frequencies
relfreq <- freq / sum(freq)
percfreq = relfreq * 100
percfreqText <- paste0(names(percfreq), ": ", round(percfreq, 2), "%")
print(percfreqText)
# 4. Generate visual outputs for the MIS dashboard
barplot(freq, main="Transactions by Payment Type", col="steelblue", las=1)
pie(freq, main="Payment Type Breakdown")
```

Exercise 2: System Performance & Quantitative Distributions

MS Excel Solution (Sturges' Rule Preparation):

1. Sample Size (n): 2000 rows.
2. Number of Classes (k): Use Sturges' formula:

$$k = 1 + 3.222 \times \log_{10}(2000) \approx 12$$

3. Class Width (\$h\$): Use =MAX(B:B) and =MIN(B:B) to find the range, then calculate the width:

$$h = \left\lceil \frac{\text{Maximum} - \text{Minimum}}{12} \right\rceil$$

R Programming Solution 1 (histogram):

```
# 1. Read your dataset and target the numeric column
listdata = read.table('omniretail_simulated_data.csv', header = TRUE, sep = ",")
numericdata <- unlist(listdata$Processing_Time)

# 2. Automate Sturges' Rule calculations
n = length(numericdata)
k = round(1 + 3.222 * log10(n), digits=0)
M = max(numericdata)
m = min(numericdata)
h = ceiling((M-m)/k)

# 3. Calculate breakpoints and distribute the data
# Note: Adding 'h' to the maximum ensures the highest value is captured in the cuts
breakpoints = seq(from = m, to = M + h, by = h)
classdata = cut(numericdata, breaks = breakpoints)

# 4. Generate the Histogram to visualize system Velocity
hist(numericdata, prob = TRUE, breaks=k, main = "Histogram of TPS Processing Times", xlab="Time
(ms)", col="lightblue")
```

R Programming Solution 2 (histogram with density curve):

```
# 1. Read your dataset and target the numeric column
listdata = read.table('omniretail_simulated_data.csv', header = TRUE, sep = ",")
numericdata <- unlist(listdata$Processing_Time)

# 2. Automate Sturges' Rule calculations
n = length(numericdata)
k = round(1 + 3.222 * log10(n), digits=0)
M = max(numericdata)
m = min(numericdata)
h = ceiling((M-m)/k)

# 3. Calculate breakpoints and distribute the data
breakpoints = seq(from = m, to = M + h, by = h)
classdata = cut(numericdata, breaks = breakpoints)

# 4. Calculate the density curve FIRST to find its maximum height
dens <- density(numericdata)
max_height <- max(dens$y)

# 5. Generate the Histogram, forcing the y-axis to fit the curve
# We use ylim = c(0, max_height * 1.1) to add a 10% blank buffer at the top
hist(numericdata, prob = TRUE, breaks=k, main = "Histogram of TPS Processing Times",
     xlab="Time (ms)", col="lightblue", ylim=c(0, max_height * 1.1))

# 6. Add the distribution (density) curve overlay
# Since we already calculated 'dens', we can just pass it directly to lines()
lines(dens, col="darkred", lwd=2)
```

Part 3: Reflection Guidelines (prepare for assessment 002)

Academic Connection & Strategic Reflection

In this workshop, you successfully managed and visualized a dataset of 2,000 records. However, the second assessment requires you to think about scale. OmniRetail Global does not process 2,000 transactions; they process millions. Use the following guidelines to draft your reflective analysis.

1. The Technical Challenge (Scaling the 5 Vs)

- Reflect on the limitations of your manual MS Excel analysis.
What happens to the *Velocity* and *Volume* of this data when the 2,000 rows you processed in milliseconds turn into 50,000 transactions per second globally?
- If OmniRetail introduces unstructured data (like customer service audio logs or social media sentiment), how will the challenge of *Variety* break the row-and-column structure you just built?

2. The Ethical & Global Challenge

- Look at the data you just generated.
It represents human behavior (payment preferences).
As an IT professional acting as a "Global Citizen", you must consider the legal and ethical implications.
- If OmniRetail scales globally, how do cross-border data management laws and data privacy regulations impact where and how this payment data is stored? What are the specific risks if this data infrastructure is outsourced?

3. Proposed Digital Management Solutions

- You must propose realistic, literature-backed solutions to the challenges you identified above. Simply stating "buy better computers" is insufficient.
- How would migrating the legacy TPS you just analyzed to a scalable Cloud Infrastructure mitigate the Velocity challenge? What specific infrastructure automation or digital management trends could ensure the system handles global data surges reliably?

Remember to support your reflection with secondary research, incorporating recent sources.

Well done!